

Student Number:.....

Teacher's Name:.....

North Sydney Boys High School



YEAR 12 Trial Higher School Certificate Examination

2002

Mathematics Extension 1

Time allowed – 2 hours (plus 5 minutes reading time)

General Instructions

- Attempt all questions on the writing paper supplied
- Write on one side of the paper only
- Start each question on a new page
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is supplied
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

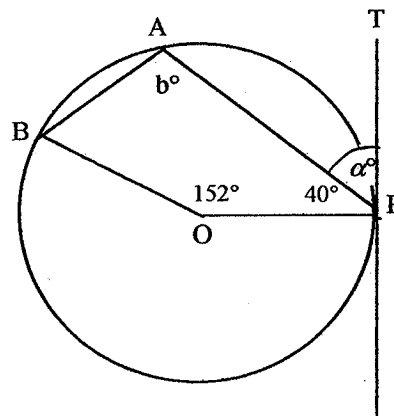
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

QUESTION 1

- (a) Differentiate: (i) $\sin^2 x$ 2
 (ii) $\sin^{-1}(2x)$ 2
- (b) Find the coordinates of the point P which divides the interval AB internally in the ratio 2 : 3 where A and B have coordinates (1, -3) and (6, 7) respectively. 2
- (c) Solve the inequality $\frac{2x+3}{x-4} > 1$ 2
- (d) $\int x\sqrt{x+1} \, dx$, using the substitution $u = 1 + x$ 3
- (e) Find $\lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{x}$ 1

QUESTION 2

- (a) PT is a tangent to the circle centre O. Find the sizes of the angles marked a and b giving reasons for your answers. 4



- (b) (i) Write down the expansion of $\tan(\alpha + \beta)$. 3
 (ii) Hence find the exact value of $\tan 75^\circ$.
- (c) Consider the function $f(x) = 3 \cos^{-1}\left(\frac{x}{2}\right)$
- (i) Evaluate $f(2)$. 1
 (ii) State the domain and range of $y = f(x)$. 2
 (iii) Draw the graph of $y = f(x)$. 2

QUESTION 3

- (a) Write $9 + 16 + 25 + \dots + n^2$ using $\sum$ notation 1
- (b) Solve $\sin 2x = \cos x$ for $0 \leq x \leq 2\pi$ 4
- (c) Find the indefinite integrals: 3
- (i) $\int \frac{dx}{x^2 + 4}$
- (ii) $\int \sin^2 2x \, dx$
- (d) Evaluate $\int_0^{\ln 3} \frac{e^x dx}{\sqrt{1+e^x}}$ using the substitution $u = e^x$. 4

QUESTION 4

- (a) The polynomial $P(x) = x^3 + ax^2 - 3ax$ has a factor $(x + 2)$.
Find the value of a . 2
- (b) Express $\sqrt{3} \cos \theta + \sin \theta$ in the form $A \cos(\theta - \alpha)$.
Hence solve the equation $\sqrt{3} \cos \theta + \sin \theta = 1$ for $-\pi \leq \theta \leq \pi$. 4
- (c) Differentiate $x \tan^{-1} x$ and hence evaluate $\int_0^1 \tan^{-1} x \, dx$. 4
- (d) Sketch $y = \sin(\cos^{-1} x)$ showing clearly the domain and range. 2

QUESTION 5

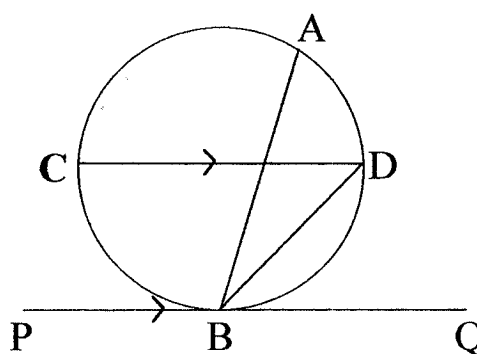
Marks

- (a) (i) Draw the graph of $y = e^{-x}$. By drawing another graph on the same set of axes, show that $f(x) = e^{-x} - x + 1$ has exactly one root. 2
- (ii) Let $x = 1$ be a first approximation to the root. Apply Newton's method once to obtain another approximation. Answer to 3 significant figures. 3
- (b) Prove that $\frac{\operatorname{cosec} \beta - \cot \beta}{\operatorname{cosec} \beta + \cot \beta} = \tan^2 \frac{\beta}{2}$. Hint: Let $\tan \frac{\beta}{2} = t$ 3

- (c) AB and CD are two intersecting chords of a circle and CD is parallel to the tangent to the circle at B. 4

Copy the diagram in your booklet.

Prove that AB bisects $\angle CAD$.



QUESTION 6

- (a) $P(4p, 2p^2)$ is any point on the parabola $x^2 = 8y$. The tangent to the parabola meets the x -axis at M, and the y -axis at N. 2
- (i) Show that the tangent at P is given by $y = px - 2p^2$. 2
- (ii) Find the co-ordinates of M and N. 2
- (iii) Find the equation of the locus of the midpoint of MN as P varies. 2
- (b) A spherical balloon leaks air such that the radius decreases at the rate of 5 mm/sec. Calculate the rate of change of the volume of the balloon when the radius is 100mm. 3
- (c) Evaluate $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$. 3

QUESTION 7

- (a) (i) Given that $S_n = 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n(n+1)$,
show by mathematical induction,

$$S_n = \frac{n}{3}(n+1)(n+2), \text{ for all positive integers } n.$$

4

- (ii) Evaluate $\lim_{n \rightarrow \infty} \frac{1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n(n+1)}{n^3}$

1

- (b) $P(x) = x^3 - 6x^2 + ax - 4$ where $a > 0$. Given that all the roots of $P(x) = 0$ are real and positive, and that one of the roots is the product of the other 2 roots, find the value of a .

3

- (c) Given $f(x) = 2\cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x^2)$. Show that $f'(x) = 0$.

4

Hence sketch $f(x)$.

End of paper

QUESTION 1

2) i) $\frac{d}{dx}(\sin^2 x) = 2 \sin x \cos x$ (2)

ii) $\frac{d}{dx} \sin^{-1} 2x = \frac{2}{\sqrt{1-4x^2}}$ (2)

b)

$P\left(\frac{3 \times 1 + 2 \times 6}{2+3}, \frac{3 \times (-3) + 2 \times 7}{2+3}\right)$
 $P(3, 1)$ (2)

c) $\frac{2x+3}{x-4} > 1$

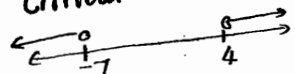
Let $\frac{2x+3}{x-4} = 1$

$2x+3 = x-4$

$x = -7$

Asymptote $x = 4$

Critical values $x = -7, 4$



$x < -7$ or $x > 4$

d) $\int x \sqrt{x+1} dx$ Let $u = 1+x$
 $du = dx$
 $= \int (u-1) \sqrt{u} du$
 $= \int u^{3/2} - u^{1/2} du$
 $= \frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} + C$
 $= \frac{2}{5} (1+x)^{5/2} - \frac{2}{3} (1+x)^{3/2} + C$ (3)

e) $\lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{\frac{x}{2}} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}}$ (1)

QUESTION 2

a) $40^\circ + a^\circ = 90^\circ$ (radius $\perp$ tangent)
 $a^\circ = 50^\circ$

Reflex $\angle$ at $O = 208^\circ$

$\therefore b^\circ = 104^\circ$ ($\angle$ at centre $= 2 \times \angle$ at circumference) (4)

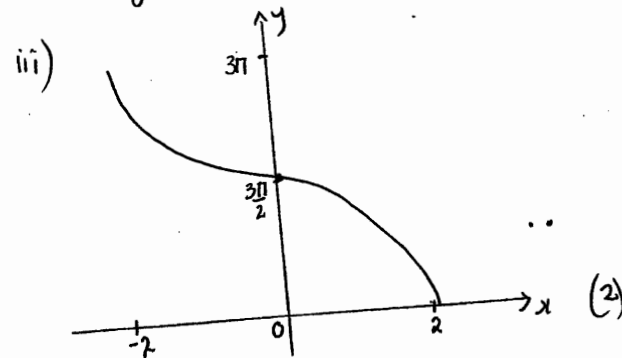
b) i) $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

ii) $\tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$
 $= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$
 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ (3)

c) $f(x) = 3 \cos^{-1} \frac{x}{2}$

i) $f(2) = 3 \cos^{-1} 1$
 $= 0$ (1)

ii) Domain $-2 \leq x \leq 2$
 Range $0 \leq y \leq 3\pi$ (2)



QUESTION 3.

$$a) 9 + 16 + 25 + \dots + n^2 = \sum_{k=3}^n k^2 \quad (1)$$

$$b) \sin 2x = \cos x$$

$$2 \sin x \cos x = \cos x$$

$$\cos x (2 \sin x - 1) = 0$$

$$\cos x = 0 \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad x = \frac{\pi}{6}, \frac{5\pi}{6} \dots$$

$$c) i) \int \frac{dx}{x^2+4} = \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C \quad (1)$$

$$ii) \int \sin^2 2x \, dx = \frac{1}{2} \int (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right] + C$$

$$= \frac{1}{2} x - \frac{1}{8} \sin 4x + C \quad (2)$$

$$d) \int_0^{\ln 3} \frac{e^x \, dx}{\sqrt{1+e^x}} = \int_1^3 \frac{du}{\sqrt{1+u}} \quad u = e^x$$

$$du = e^x \, dx$$

$$= \int_1^3 (1+u)^{-1/2} \, du \quad \text{When } x=0, u=1$$

$$= 2 \left[(1+u)^{1/2} \right]_1^3 \quad x=\ln 3, u=3$$

$$= 2 [2 - \sqrt{2}]$$

$$= 4 - 2\sqrt{2} \quad (4)$$

QUESTION 4

$$a) P(x) = x^3 + ax^2 - 3ax$$

$$\text{If } x+2 \text{ is factor } P(-2) = 0$$

$$\therefore -8 + 4a + 6a = 0$$

$$10a = 8$$

$$a = \frac{4}{5} \quad (2)$$

$$b) \sqrt{3} \cos \theta + \sin \theta = A \cos(\theta - \alpha)$$

$$A \cos(\theta - \alpha) = A \cos \theta \cos \alpha + A \sin \theta \sin \alpha$$

$$\therefore A \cos \alpha = \sqrt{3}$$

$$A \sin \alpha = 1$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$A^2 = 1+3$$

$$A = 2$$

$$\therefore \sqrt{3} \cos \theta + \sin \theta = 2 \cos(\theta - \frac{\pi}{6})$$

$$2 \cos(\theta - \frac{\pi}{6}) = 1$$

$$\cos(\theta - \frac{\pi}{6}) = \frac{1}{2}$$

$$\theta - \frac{\pi}{6} = -\frac{\pi}{3}, \frac{\pi}{3}$$

$$\theta = -\frac{\pi}{6}, \frac{\pi}{2}$$

(4)

$$c) y = x \tan^{-1} x$$

$$\frac{dy}{dx} = 1 \cdot \tan^{-1} x + x \cdot \frac{1}{1+x^2}$$

$$\left[x \tan^{-1} x \right]' = \int \tan^{-1} x \, dx + \int \frac{x}{1+x^2} \, dx$$

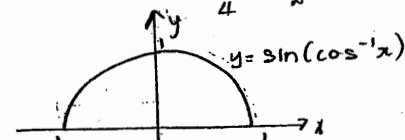
$$\therefore \int \tan^{-1} x \, dx = \left[x \tan^{-1} x \right]' - \int \frac{x}{1+x^2} \, dx$$

$$= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]'$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

(4)

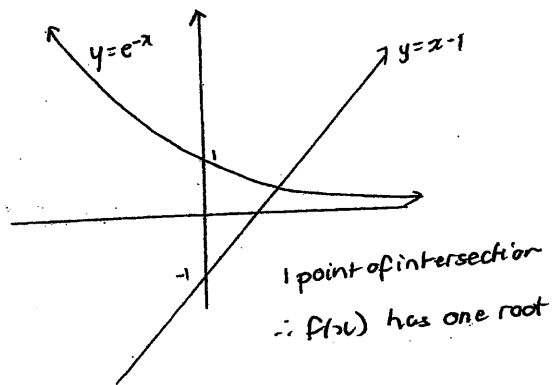
d)



(2)

Question 5

a) i) $f(x) = e^{-x} - x + 1 = 0$
 $e^{-x} = x - 1$



(2)

ii) $f(x) = e^{-x} - x + 1$, $f(1) = e^{-1} \approx 0.3679$
 $f'(x) = -e^{-x} - 1$, $f'(1) = -e^{-1} - 1 \approx -1.3679$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

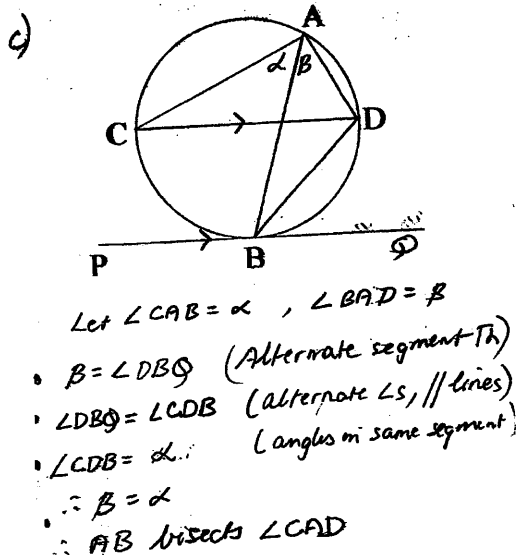
$$= 1 - \frac{e^{-1}}{-e^{-1} - 1}$$

$$\approx 1.27$$

(3)

i) $\tan \frac{B}{2} = t$
 L.H.S = $\frac{\operatorname{cosec} B - \cot B}{\operatorname{cosec} B + \cot B}$
 $= \frac{\frac{1+t^2}{2t} - \frac{1-t^2}{2t}}{\frac{1+t^2}{2t} + \frac{1-t^2}{2t}}$
 $= \frac{1+t^2 - 1+t^2}{1+t^2 + 1-t^2}$
 $= \frac{2t^2}{2}$
 $= t^2$
 $= \tan^2 \frac{B}{2}$
 $= \text{R.H.S}$

(3)



(4)

Question 6

i) $x^2 = 8y$
 $y = \frac{x^2}{8}$
 $\frac{dy}{dx} = \frac{x}{4}$

at $P(4p, 2p^2)$, $m = p$
 eqn. of tangent: $y - y_1 = m(x - x_1)$
 $y - 2p^2 = p(x - 4p)$
 $\therefore y = px - 2p^2$ (2)

ii) X-axis intercept: $y = 0$

$\therefore px = 2p^2$
 $x = 2p$

$\therefore M(2p, 0)$

Y-axis intercept: $x = 0$

$\therefore y = -2p^2$

$N(0, -2p^2)$

iii) Midpoint of $MN = \left(\frac{2p+0}{2}, \frac{0-2p^2}{2}\right)$

i.e. $(p, -p^2)$

$x = p$

$y = -p^2$

$\therefore y = -x^2$ is locus. (2)

$V = \frac{4}{3}\pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dr}{dt} = -5 \text{ mm/sec}$

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$= 4\pi r^2 \times -5$

$= -20\pi r^2$

when $r = 100 \text{ mm} = 10 \text{ cm}$

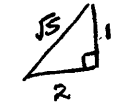
$\frac{dV}{dr} = -200000\pi \text{ mm}^2/\text{sec}$

or decreasing at a rate of $200000\pi \text{ mm}^2/\text{sec}$

c) Let $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = x$

Let $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) = \alpha$

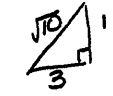
$\sin \alpha = \frac{1}{\sqrt{5}}$



$\therefore \cos \alpha = \frac{2}{\sqrt{5}}$

Let $\sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = \beta$

$\sin \beta = \frac{1}{\sqrt{10}}$



$\cos \beta = \frac{3}{\sqrt{10}}$

$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
 $= \frac{1}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}}$

$= \frac{5}{\sqrt{50}}$

$= \frac{1}{\sqrt{2}}$

$\therefore \alpha + \beta = \frac{\pi}{4}$

$\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = \frac{\pi}{4}$

(3)

1/9

(3)

QUESTION 7

a) i) $S_n = 1 \times 2 + 2 \times 3 + \dots + n(n+1) = \frac{n}{3} (n+1) (n+2)$

① If $n=1$

L.H.S = $1 \times 2 = 2$

R.H.S = $\frac{1}{3} (1+1) (1+2) = 2$

$\therefore$ True for $n=1$

② Assume true for $n=k$

i.e. $1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k}{3} (k+1) (k+2)$

③ If $n=k+1$

$S_{k+1} = 1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2)$

$= \frac{k}{3} (k+1) (k+2) + (k+1) (k+2)$

$= \frac{(k+1) (k+2) (k+3)}{3}$

$\therefore$ True for $n=k+1$

④ Since result is true for $n=1$, it is true for next integer, $n=2$ (i.e. $n+1$) and so it is true for $n=3$ and so on. $\therefore$ True for all integers n .

ii) $\lim_{n \rightarrow \infty} \frac{1 \times 2 + 2 \times 3 + \dots + n(n+1)}{n^3} = \lim_{n \rightarrow \infty} \frac{\frac{n}{3} (n+1) (n+2)}{n^3}$
 $= \lim_{n \rightarrow \infty} \frac{\frac{1}{3} (1 + \frac{1}{n}) (1 + \frac{2}{n})}{1}$
 $= \frac{1}{3}$

b) $P(x) = x^3 - 6x^2 + ax - 4$

Let roots be $\alpha, \beta, \alpha\beta$

$\alpha + \beta + \alpha\beta = 6$ — (1)

$\alpha\beta + \alpha^2\beta + \alpha\beta^2 = a$ — (2)

$\alpha^2\beta^2 = 4$ — (3)

$\alpha\beta = \pm 2$

$\alpha\beta = 2$ (since roots are +ve).

(1): $\alpha + \beta + 2 = 6$
 $\alpha + \beta = 4$

(2): $\alpha\beta(1 + \alpha + \beta) = a$

$\therefore 2(1 + 4) = a$

$\therefore a = 10$

(3)

7 c) $f(x) = 2 \cos^{-1} \left(\frac{x}{\sqrt{2}} \right) - \sin^{-1} (1 - x^2)$

$f'(x) = -\frac{2}{\sqrt{2}} \frac{1}{\sqrt{1 - \frac{x^2}{2}}} - \frac{-2x}{\sqrt{1 - (1 - x^2)^2}}$

$= \frac{-2}{\sqrt{2 - x^2}} + \frac{2x}{\sqrt{1 - (1 - 2x^2 + x^4)}}$

$= \frac{-2}{\sqrt{2 - x^2}} + \frac{2x}{\sqrt{2x^2 - x^4}}$

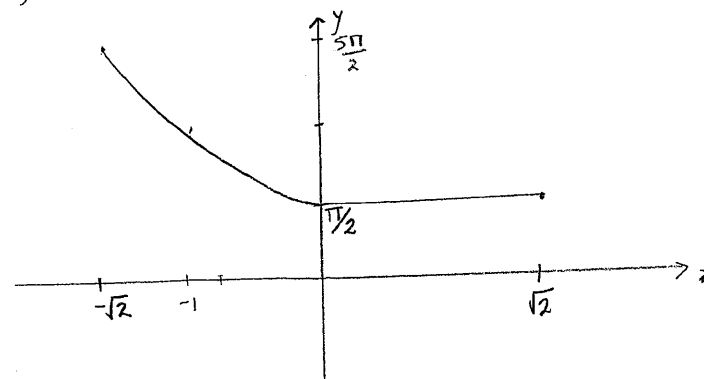
$= \frac{-2}{\sqrt{2 - x^2}} + \frac{2}{\sqrt{2 - x^2}} \quad \text{if } x > 0, \sqrt{x^2} = |x| = x$

$= 0 \quad \text{if } x > 0 \quad \left(f'(x) = \frac{-4}{\sqrt{2 - x^2}} \quad \text{if } x < 0 \right)$

If $f'(x) = 0$, $f(x)$ is constant

Domain: $-1 \leq \frac{x}{\sqrt{2}} \leq 1 \quad \times \quad -1 \leq 1 - x^2 \leq 1$
 $-\sqrt{2} \leq x \leq \sqrt{2} \quad \quad \quad -\sqrt{2} \leq x \leq \sqrt{2}$

$f(0) = 2 \times \cos^{-1}(0) - \sin^{-1}(1) = \frac{\pi}{2}$



$f'(x) = 0$

- GRAPH
- CORRECT DOMAIN
- $f(x) = \text{constant} \quad (x > 0)$
- $f(x) = \frac{\pi}{2}$

(4)